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Magnetization process under generically directed field in GO Fe–(3 wt%)Si laminations

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Abstract

It is shown that magnetization curve and hysteresis loops in any direction of grain-oriented Fe–Si laminations can be predicted from the experimental knowledge of curve and loops taken along the rolling and the transverse directions, with no adjustable or arbitrary parameters. This conclusion follows from the physical analysis of the involved magnetization processes: displacement of the 180° domain walls belonging to the [0 0 1] easy axis, transition of the [0 0 1] oriented domains into a balanced system of [1 0 0] and [0 $\bar{1}$ 0] directed domains and vice versa, coherent rotations. Results and domain observations concerning high permeability Fe–(3 wt%)Si strip and sheet samples are reported and discussed here.

Keywords: Grain-oriented steel; Magnetization processes; Magnetization curves; Hysteresis loop

Grain-oriented Fe–(3 wt%)Si laminations owe their continuing importance as a unique subject of basic research studies and industrial applications to their excellent crystallographic properties. This makes them prone to reasonable magnetic modeling and, at the same time, an ideal material for many efficient electromagnetic devices. While in many cases physical modeling has been limited, according to the prevailing applicative requirements, to the properties related to the rolling direction (RD), the use of computational methods in magnetic cores increasingly calls for the knowledge, at least approximate, of magnetization curve and hysteresis loops in directions different from RD [1]. However, the possibility of defining an intrinsic magnetization behavior in a [0 0 1](1 1 0) Fe–Si single crystal is limited to the [0 0 1] and 0$\bar{1}$0 directions, since the magnetization process in all the other directions is affected by the sample geometry. It has actually been suggested that the magnetic properties along RD ($\sim$[0 0 1]) and the transverse direction (TD, $\sim$[0 $\bar{1}$ 0]) in GO laminations can be exploited in trying to assess the material behavior along the other directions [1–3]. The proposed models, however, treat the material as a continuum and eventually overlook the non-intrinsic character of the magnetization curves aimed at. In the present paper, we show that it is possible to understand and predict hysteresis loops and magnetization curve in GO laminations for a generically directed applied field in terms of RD and TD properties, through direct consideration of the involved domain wall (d.w.) processes and their dependence on the measuring configuration [4]. Quasi-static hysteresis loops, energy losses and normal magnetization curve have been measured in 0.30 mm thick high permeability GO Fe–(3 wt%)Si narrow (Epstein, 30 mm × 305 mm) and wide (120 mm × 305 mm) strips, cut along directions ranging between RD and TD at 15° intervals. The narrow strips have been tested in a conventional Epstein frame and the wide strips in a double-C laminated yoke, either as individual (SST) or cross-stacked sheets. With the Epstein frame and parallel stacking we obtain a condition of high lateral demagnetizing coefficient and near-zero transverse magnetization component $\boldsymbol{J}_{\perp}$. The total magnetization $\boldsymbol{J}$ and the longitudinal magnetization $\boldsymbol{J}_{\parallel}$ practically coincide in this case. With cross-stacked wide strips (X-stacking) the demagnetizing fields at the strip edges are very small and the ideal condition of infinitely extended sample is emulated [5]. An

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intermediate condition is reached by means of the SST configuration. The domain structure and its evolution with the applied field have been directly observed by means of a longitudinal Kerr-effect set-up. Fig. 1 provides an example of domain observation along the magnetization curve. The images refer to four different $\boldsymbol{J}$ levels, simultaneously considered in Epstein and X-stacked strips (cutting angle $\theta = 75°$). Starting from the demagnetized state, where the domain structure is invariably made of a system of antiparallel domains with $\boldsymbol{J}_s$ (saturation magnetization) directed along [0 0 1] and [0 0 $\bar{1}$], three different processes take place: (1) displacement of 180° domain walls (d.w.s.) ([0 0 1] and [0 0 $\bar{1}$] phases); (2) balanced growth of the 90° [1 0 0] and [0 $\bar{1}$ 0] phases out of the previous 180° ones, with no net magnetization normal to the sheet plane (chevron-type and transverse flux-closing domain structures); (3) coherent rotations towards the applied field direction. We can see in Fig. 1 that process (1) combines with process (2) in the Epstein sample, while in the X-stacked samples it is basically concluded when process (2) starts. This is in qualitative agreement with the correspondingly observed hysteresis loops. It is then understood that the domain processes occurring in the RD and TD cut strips, reasonably assimilated to [0 0 1] and [0 $\bar{1}$ 0] oriented single crystal strips, are (1) and (2), respectively. The macroscopic magnetization value has an obvious relationship with the domain structure in the RD cut strips, while it is proportional to the fractional volume v_{90} occupied by the 90° phases in the TD cut strips, according to the expression $J_{90}^{(r)} = (J_s/\sqrt{2}) \cdot v_{90}$. The RD and TD hysteresis loops, intrinsic in nature, are assumed to play the role of building blocks of the loops along a generic orientation in the lamination plane. They contribute in proportions determined by the value of the components of the internal field $\boldsymbol{H}_i = \boldsymbol{H} + \boldsymbol{H}_d$ (vectorial sum of applied field $\boldsymbol{H}$ and demagnetizing field $\boldsymbol{H}_d$) along RD and TD, respectively. The domain processes cover the so-called *mode I* in the framework of Néel's phase theory [6]. The rotations cover *modes II–IV* and the corresponding magnetization curves can in principle be calculated by applying a minimum free energy principle. While for the RD applied field only *mode I* exists (two phases), the TD process follows the sequence: *mode I* (four phases), *mode III* (two phases).

With these premises, the calculation of the hysteresis loop along a generic direction of the GO lamination (i.e. for a generic cutting angle θ in a strip sample) for a given periodic applied field $H(t)$ can be carried out according to the procedure summarized in the following.

1. *Narrow strips (Epstein)*. As previously stressed and as verified by the experiments, in this case $\boldsymbol{J}_\perp \sim 0$. For

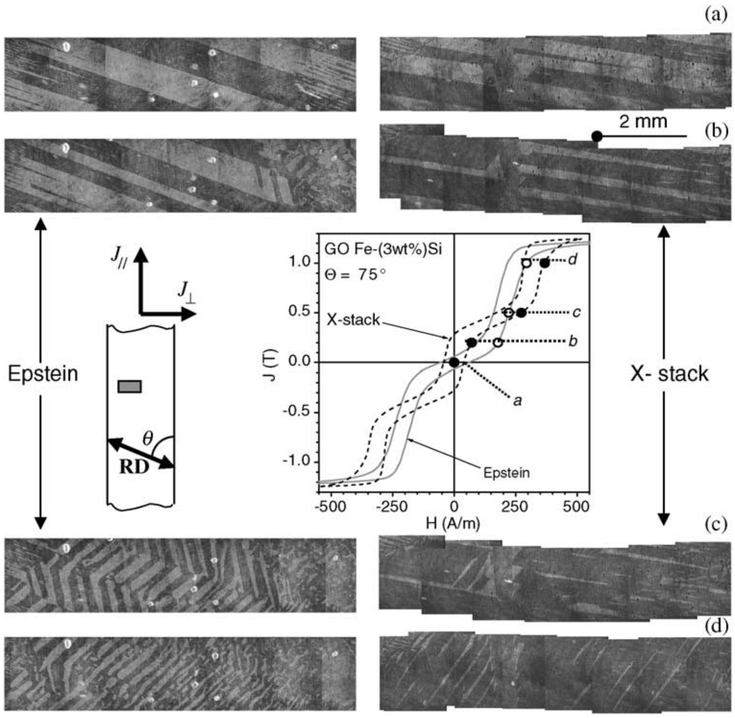

Fig. 1. Domain structure versus magnetization in Epstein strips (open symbols) and cross-stacked wide strips (full symbols) cut at an angle $\theta = 75°$ to the rolling direction (RD). The different course (*a–d*) followed in the growth of the [1 0 0] and [0 $\bar{1}$ 0] phases out of the [0 0 1] directed antiparallel domain structure characterizing the demagnetized state is related to the presence (Epstein) or absence (X-stack) of the demagnetizing field at the strip edges.

given volumic fractions v_{90} and $v_{180} = 1 - v_{90}$ of the 90° and 180° phases, we define the corresponding magnetizations, directed along RD and TD, respectively, as $\boldsymbol{J}_{90}$ and $\boldsymbol{J}_{180}$. They are related to $\boldsymbol{J}_{\parallel} \sim \boldsymbol{J}$ and $\boldsymbol{J}_{\perp}$ by the expressions

$$J_{\parallel} = J_{180} \cos\theta + J_{90} \sin\theta = J,$$
$$J_{\perp} = J_{180} \sin\theta - J_{90} \cos\theta = 0 \quad (1)$$

from which we obtain $J_{180} = J\cos\theta$ and $J_{90} = J\sin\theta$. If referred to the volumic fractions v_{90} and v_{180}, they become $J^{(r)}_{180} = J_{180}/v_{180}$ and $J^{(r)}_{90} = J_s/\sqrt{2}$. We assume that the RD and TD hysteresis loops of peak magnetization value $J^{(r)}_{180p}$ and J_{90p}, parametrically expressed versus time as $(H_{180}(t), J^{(n)}_{180}(t)), (H_{90}(t), J_{90}(t))$, are known by experiments. $H_{180}(t)$ and $H_{90}(t)$ are related to the longitudinal applied field $H(t)$ and the transverse demagnetizing field $H_d(t)$ by the equations

$$H_{180}(t) = H(t)\cos\theta - H_d(t)\sin\theta,$$
$$H_{90}(t) = H(t)\sin\theta + H_d(t)\cos\theta \quad (2)$$

which provide

$$H(t) = H_{180}(t)\cos\theta + H_{90}(t)\sin\theta,$$
$$H_d(t) = H_{90}(t)\cos\theta - H_{180}(t)\sin\theta. \quad (3)$$

It is then concluded that, if we impose a macroscopic $\boldsymbol{J}(t)$ behavior, we are able to reconstruct by Eqs. (1)–(3) the corresponding $\boldsymbol{H}(t)$ function and consequently achieve the desired hysteresis loop by composition of RD and TD loops. Obviously, t enters these equations as a parameter and no dynamic effects are considered. It is also clear that the hysteresis loop prediction includes that of the energy loss and of the normal magnetization curve. The latter can be predicted further when the magnetization process enters the rotational regime and *mode II* unfolds, starting around the ideal remanence $J_{I-II} = J_s/(l+m+n)$, where l, m, n are the direction cosines made by $\boldsymbol{H}$ with the quaternary axis $\langle 1\,0\,0 \rangle$ [6].

2. *X-stacked laminations*. With this configuration $\boldsymbol{H}_d \sim 0$ and $\boldsymbol{J}_{\perp} \neq 0$. We are again interested in the $(\boldsymbol{H}, \boldsymbol{J}_{\parallel})$ curve. If $\theta < 54.7°$, *mode I* takes place by unbalance of the 180° phases only (i.e. by 180° d.w. displacements), because the 90° phases are energetically unfavored ($v_{180} = 1$). The measured $(H(t), J_{\parallel}(t))$ loop is obtained from the $(H_{180}(t), J_{180}(t))$ loop by simple projection, with $H_{180}(t) = H(t)\cos\theta$ and $J(t) = J_{180}(t)\cos\theta$. *Mode I* ends when the 180° phases become saturated and is followed by *Mode IV*, where the [0 0 1] directed magnetization rotates in the (1 1 0) plane towards the direction of $\boldsymbol{H}$ [4]. If $\theta > 54.7°$, all four phases can coexist in *mode I*, but the 180° phases are still favored in the demagnetized state. An applied field $H(t)$ has components along RD and TD $H_{180}(t) = H(t)\cos\theta$ and $H_{90}(t) = H(t)\sin\theta$, respectively, and the corresponding magnetization values $J_{90}(t)$ and $J^{(r)}_{180}(t)$ are known. The longitudinal magnetization $J_{\parallel}(t)$ is easily recovered by means of Eq. (1), where $J_{180}(t) = J^{(r)}_{180}(t) \cdot v_{180}(t)$, and the loop $((H(t), J_{\parallel}(t))$ is eventually obtained. By this procedure, a prediction like the one shown in Fig. 2 is obtained ($\theta = 75°, J_p = 1.15$ T). Note how the dramatic difference between the Epstein and the X-stack

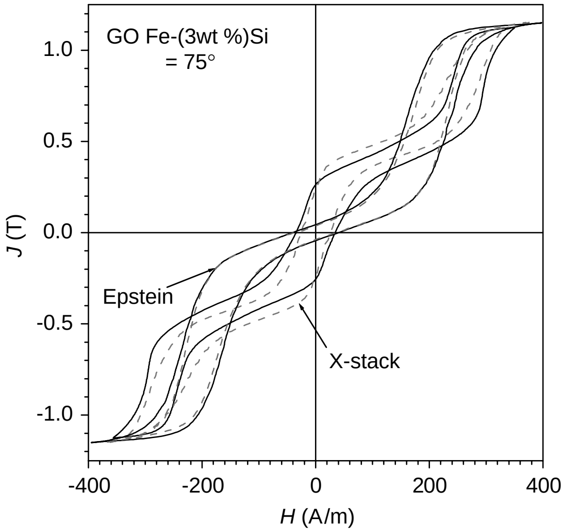

Fig. 2. Hysteresis loops measured on GO Epstein strips and 120 mm wide X-stacked sheets (solid lines), cut at an angle $\theta = 75°$ with respect to RD, and their theoretical predictions (dashed lines).

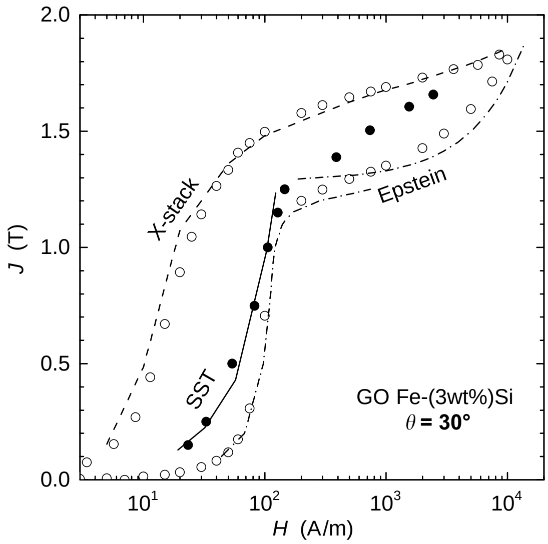

Fig. 3. Experimental (symbols) and theoretical (lines) normal magnetization curves for sheet cutting angle $\theta = 30°$ under three measuring configurations. The SST experiments have been carried out on 120 mm × 300 mm single sheets, magnetized along their length and flux closed by means of a double-C laminated yoke.

loops can be handled by the model. *Mode III*, where the magnetization vectors belonging to the [1 0 0] and [0 $\bar{1}$ 0] axes symmetrically rotate towards the direction of $\boldsymbol{H}$, follows *mode I.* The angle to be covered by the rotation to reach saturation is $\varphi = \cos^{-1}(\sin\theta/\sqrt{2})$ and the corresponding curve can again be obtained by minimizing the sum of anisotropy and Zeeman energies [4].

2. *General case*. For a wide strip, a square or a circular sample additional knowledge of the magnetometric demagnetizing coefficient N is required, in order to explicitly express $H_{\rm d}(t) = (N/\mu_0)\cdot J_\perp(t)$ in Eq. (2). The RD and TD loops must have peak field and peak magnetization values satisfying the equation

$$\left[H_{180\rm p} + \frac{N}{\mu_0} v_{180\rm p} J^{(r)}_{180\rm p}\right]\sin\theta = \left[H_{90\rm p} + \frac{N}{\mu_o} J_{90\rm p}\right]\cos\theta, \quad (4)$$

where $J(t)$ and $H(t)$ are again given by Eqs. (1) and (3), the latter being the internal field in the given direction (i.e. the field measured by means of an H-coil) if the measurement is performed on an open sample. Fig. 3 provides an example of prediction of normal magnetization curves for X-stack, narrow-strip (Epstein), and wide-strip (SST) samples. In the latter case, the prediction is limited to the lower part of the curve (*mode I*), because the rotating *modes* lack the specific constraints exhibited by the other two cases and cannot be handled in a simple way. Some kind of interpolation between the Epstein and X-stack curves might, however, be considered.

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